

E-COMMERCE SALES ANALYTICS DASHBOARD USING PYTHON

PROJECT REPORT

Submitted as part of IBM SkillsBuild Academic Internship

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1. Abstract

This project presents an E-Commerce Sales Analytics Dashboard developed using Python. The purpose is to convert raw sales transaction data into meaningful business insights through data preparation, analysis, visualization, and interactive reporting. The dashboard uses a Superstore sales dataset containing 10,800 records and 21 columns. Key performance indicators such as Total Sales, Total Profit, Total Quantity, and Total Orders are displayed along with charts for category, region, product, customer segment, monthly sales, discounts, and profitability. Streamlit is used to build the interactive dashboard, while Pandas and Plotly are used for data analysis and visualization.

2. Introduction

E-commerce businesses generate large amounts of sales data. Simply storing this data is not sufficient; it must be analyzed to understand business performance. Sales analytics helps organizations identify trends, understand customer segments, compare products and regions, and monitor profitability. This project develops a simple interactive dashboard that presents these insights in an easy-to-understand form.

3. Problem Statement

Raw e-commerce sales data contains many records and fields, making it difficult to identify important business patterns manually. There is a need for an interactive system that can summarize sales performance, compare products and regions, analyze customers and profitability, and present useful insights through visualizations.

4. Objectives

- Load and analyze e-commerce sales transaction data using Python.
- Prepare and organize the dataset for analysis.
- Calculate important sales and profitability KPIs.
- Analyze sales by category, sub-category, region, product, and customer segment.
- Identify monthly sales trends and loss-making areas.
- Study the relationship between discounts and profit.
- Build an interactive dashboard using Streamlit.
- Present business insights and practical recommendations.

5. Dataset Description

The project uses a Superstore sales dataset containing 10,800 sales transaction records and 21 columns. It contains information about orders, customers, products, locations, sales, quantity, discount, and profit.

Property	Details
Records	10,800
Columns	21
Format	CSV
Main measures	Sales, Quantity, Discount, Profit

Main dimensions	Category, Sub-Category, Region, Segment, Product
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6. Technologies Used

Technology	Purpose
Python	Main programming language
Pandas	Data loading, transformation, and analysis
NumPy	Numerical processing support
Plotly	Interactive charts and visualization
Streamlit	Interactive dashboard
VS Code	Development environment
Git/GitHub	Version control and project sharing

7. Methodology

- Data Collection: Obtain the Superstore CSV dataset.
- Data Preparation: Load the dataset using Pandas and convert date columns.
- Data Analysis: Calculate KPIs and group data by category, region, product, segment, and time.
- Visualization: Create interactive charts using Plotly.
- Dashboard Development: Display KPIs, charts, filters, insights, and recommendations using Streamlit.
- Business Interpretation: Convert analysis results into useful insights and actions.

8. System / Project Workflow

Superstore.csv → Pandas Data Loading → Data Preparation → KPI Calculation → Data Analysis → Plotly Visualization → Streamlit Dashboard → Business Insights

The project does not require a separate backend server or database. The CSV file acts as the data source, Python performs the analysis, Plotly creates visualizations, and Streamlit provides the dashboard interface.

9. Dashboard Modules

9.1 Executive Overview

Displays Total Sales, Total Profit, Total Quantity, and Total Orders for a quick business summary.

9.2 Sales Analysis

Includes Sales by Category, Monthly Sales Trend, and Sales by Region to identify important sales patterns.

9.3 Product Analysis

Includes Sales by Sub-Category and Top 10 Products by Sales.

9.4 Customer and Segment Analysis

Shows sales distribution across customer segments and customer counts by region.

9.5 Profit and Risk Analysis

Compares profit across categories, identifies loss-making sub-categories, and includes a Discount vs Profit scatter plot.

9.6 Interactive Filters

Users can filter the dashboard by Region and Category. KPI values and charts update according to the selections.

9.7 Business Insights and Recommendations

- Focus marketing efforts on high-performing categories and regions.
- Review loss-making sub-categories and products.
- Monitor discounts carefully because excessive discounts can reduce profitability.
- Study high-value customer segments and develop retention strategies.
- Use monthly sales trends for inventory and promotional planning.

10. Results

The completed dashboard successfully loads the sales dataset and provides interactive analysis through KPI cards, charts, filters, and business insights. Users can select different regions and categories and observe the corresponding changes. The project demonstrates how Python-based data analytics can transform raw transaction data into a simple business intelligence dashboard.

11. Advantages

- Simple and easy-to-use interface.
- Interactive filtering by region and category.
- Multiple business KPIs in one dashboard.
- Interactive Plotly visualizations.
- Sales, product, customer, and profitability analysis.
- Easy to demonstrate using Streamlit.

12. Limitations

- The project uses a static CSV dataset.
- It does not use a live e-commerce database.
- Recommendations are based on historical data patterns.
- Advanced forecasting and machine learning are not included.
- User authentication is not implemented.

13. Future Enhancements

- Connect the dashboard to a live database.
- Add sales forecasting.
- Add advanced customer analysis.

- Add downloadable reports and filtered datasets.
- Add more KPIs and date filters.
- Deploy the dashboard online.

14. Project Structure

The following structure represents the project files shared in the GitHub repository:

E-Commerce-Sales-Analytics/

```

■■■ app.py
■■■ Superstore.csv
■■■ requirements.txt
■■■ README.md

```

Note: .venv is used only as a local Python virtual environment on the development computer and is not uploaded to GitHub.

15. How to Run the Project

- Open the project folder in VS Code.
- Activate the Python virtual environment.
- Install packages using requirements.txt.
- Make sure Superstore.csv is in the project folder.
- Run: streamlit run app.py
- Open the local Streamlit address shown in the terminal.

16. References

- Superstore sales dataset used for academic data analysis.
- Python documentation.
- Pandas documentation.
- Plotly documentation.
- Streamlit documentation.
- IBM SkillsBuild Academic Internship learning materials.

17. Conclusion

The E-Commerce Sales Analytics Dashboard demonstrates a complete data analytics workflow using Python. The project transforms raw sales transactions into KPIs, visualizations, insights, and recommendations. Pandas is used for analysis, Plotly for visualization, and Streamlit for the dashboard. The project provides a foundation for future enhancements such as live databases, forecasting, advanced customer analytics, and deployment.

Appendix A – Dashboard Screenshots

The following appendix includes the complete dashboard screenshot sequence from the submitted dashboard PDF. All relevant views are retained so that no chart or section is omitted.

Dashboard View 1: Executive Overview

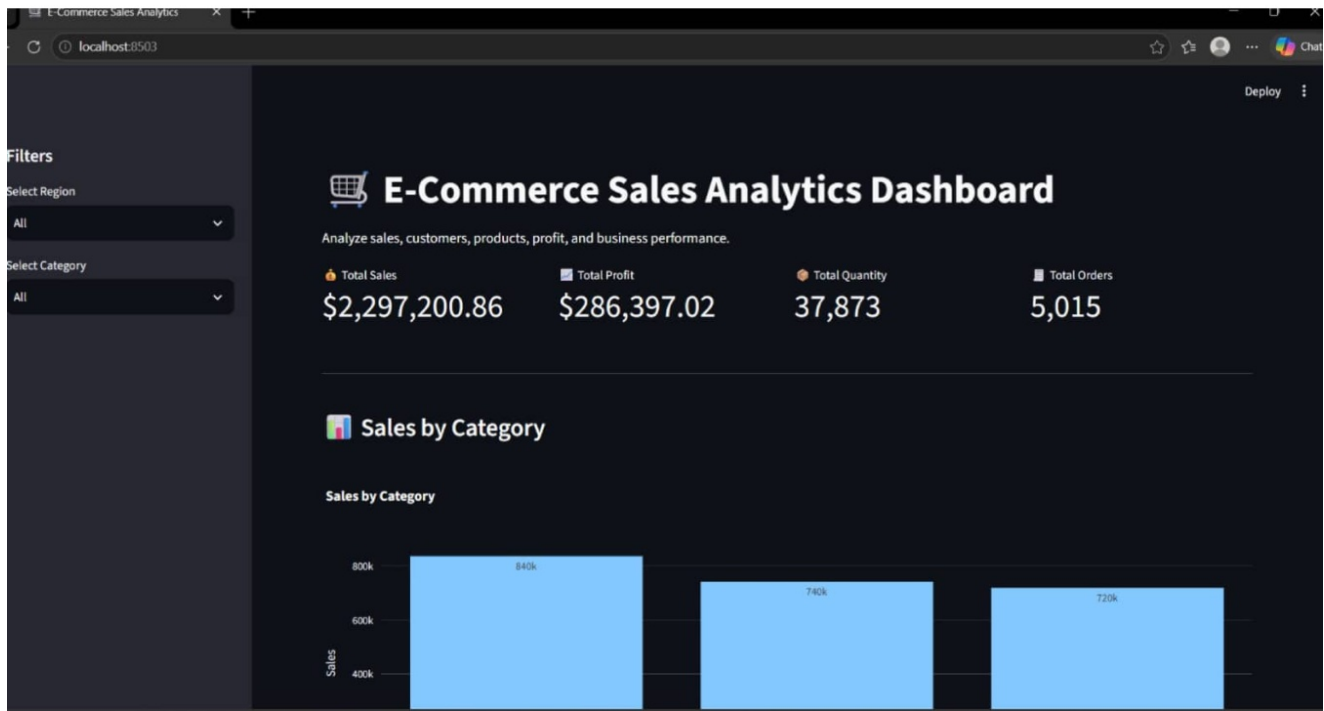


Figure A1: Executive Overview – dashboard view 1.

Dashboard View 2: Sales Analysis

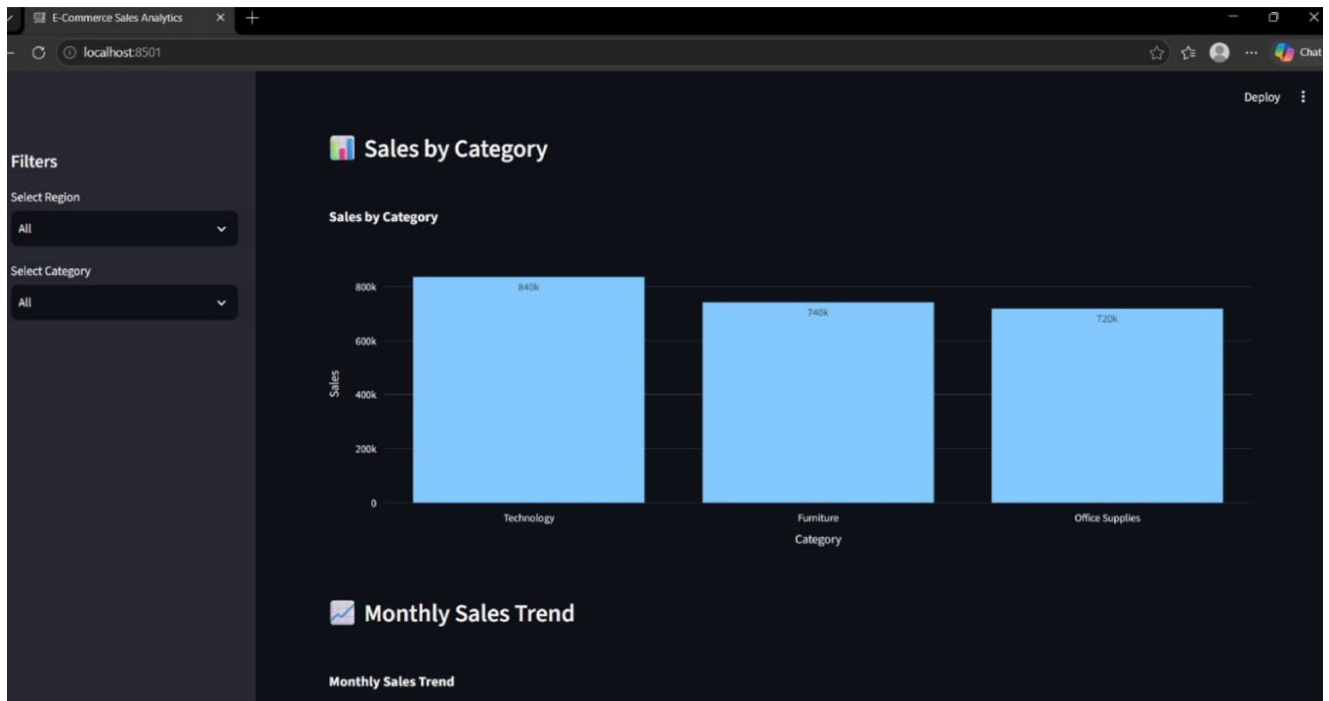


Figure A2: Sales Analysis – dashboard view 2.

Dashboard View 3: Sales Analysis



Figure A3: Sales Analysis – dashboard view 3.

Dashboard View 4: Sales Analysis

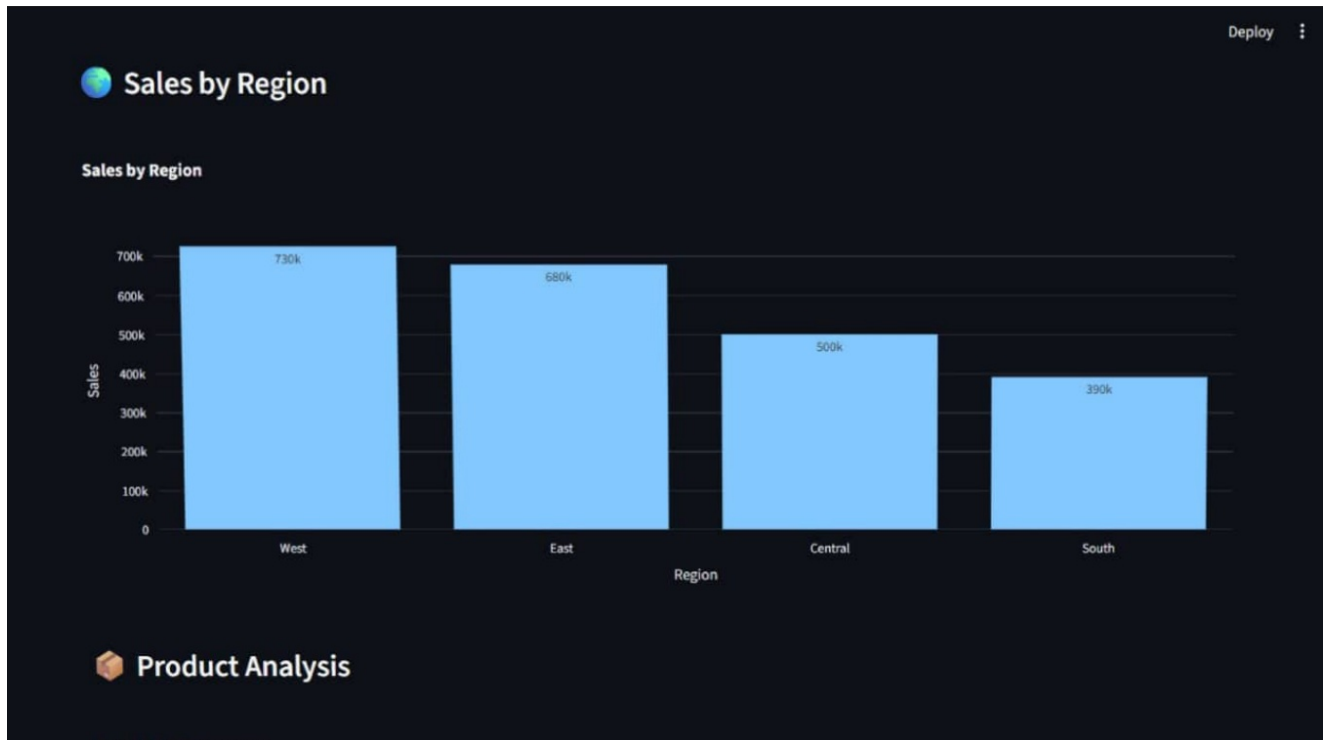


Figure A4: Sales Analysis – dashboard view 4.

Dashboard View 5: Product Analysis

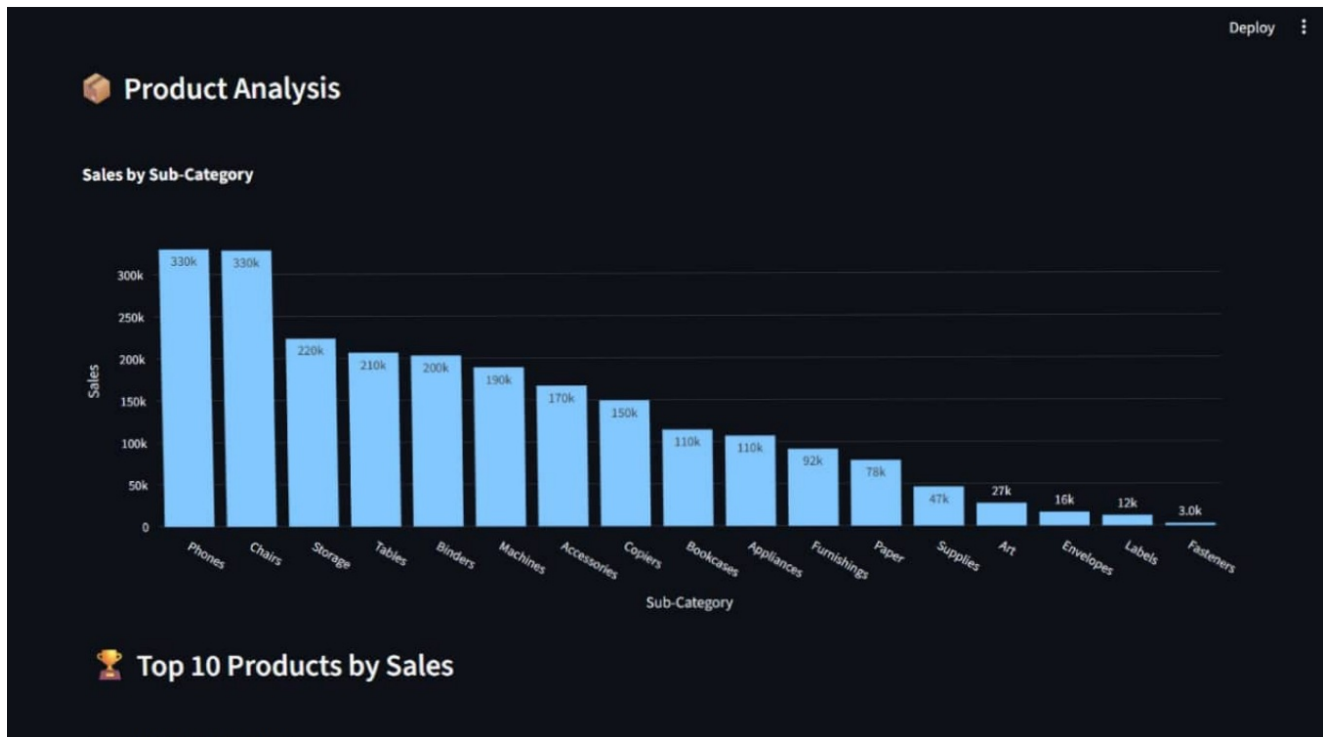


Figure A5: Product Analysis – dashboard view 5.

Dashboard View 6: Product Analysis

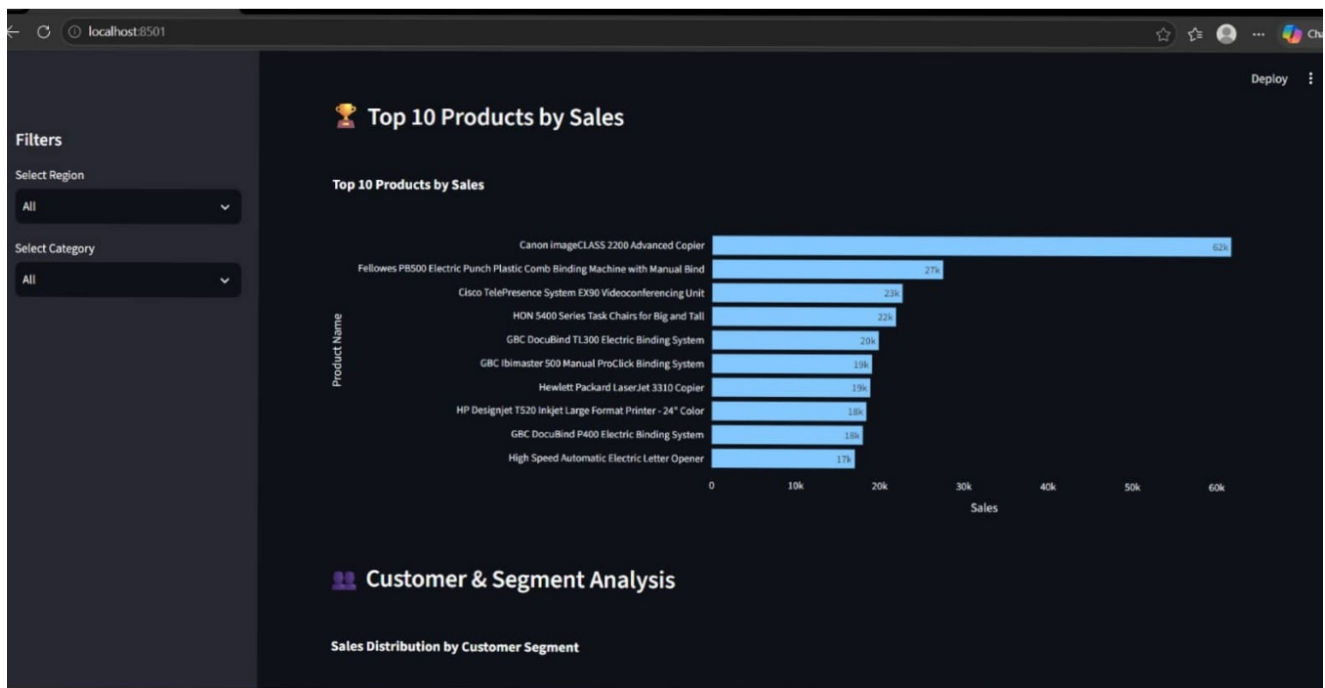


Figure A6: Product Analysis – dashboard view 6.

Dashboard View 7: Customer and Segment Analysis

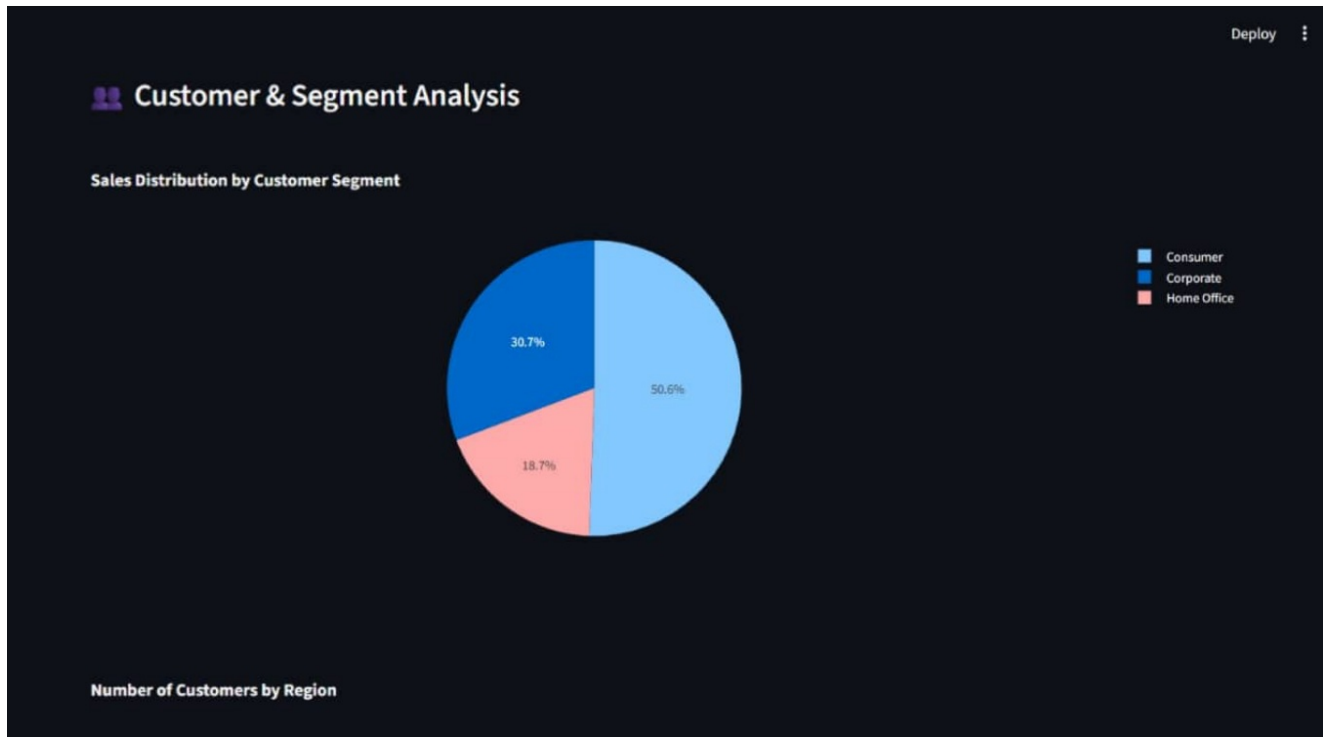


Figure A7: Customer and Segment Analysis – dashboard view 7.

Dashboard View 8: Customer and Segment Analysis

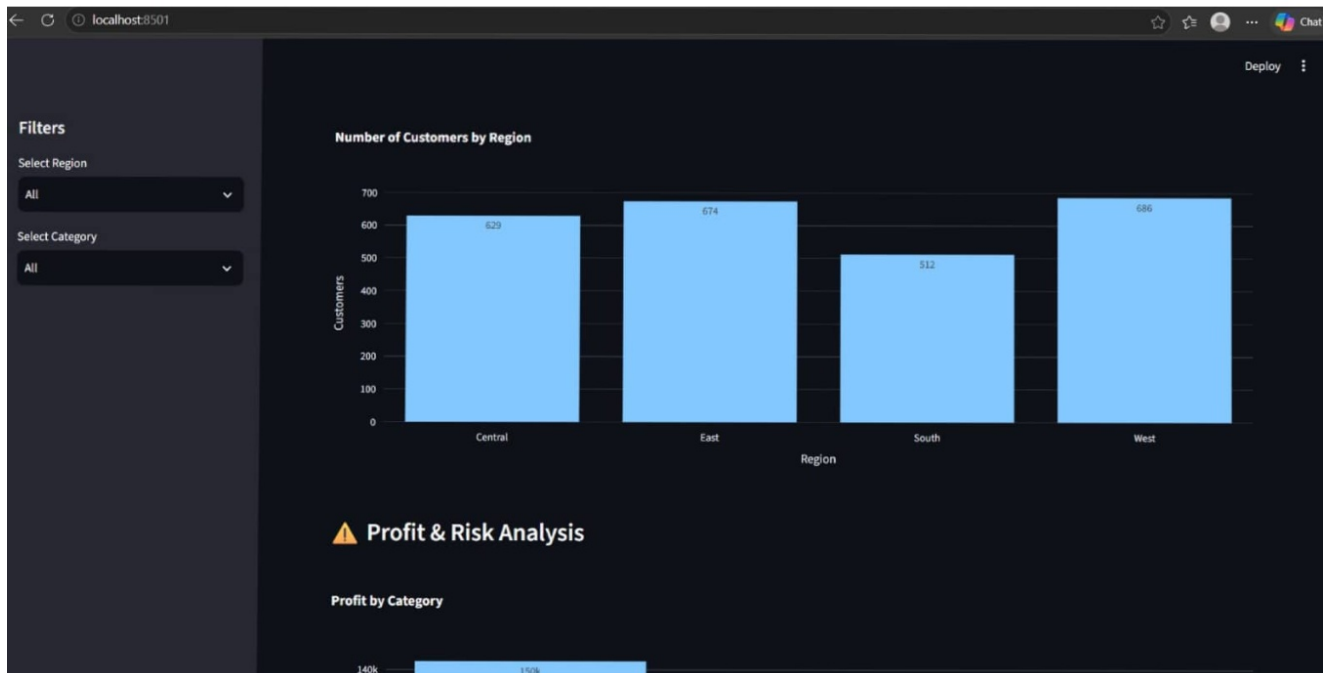


Figure A8: Customer and Segment Analysis – dashboard view 8.

Dashboard View 9: Profit and Risk Analysis

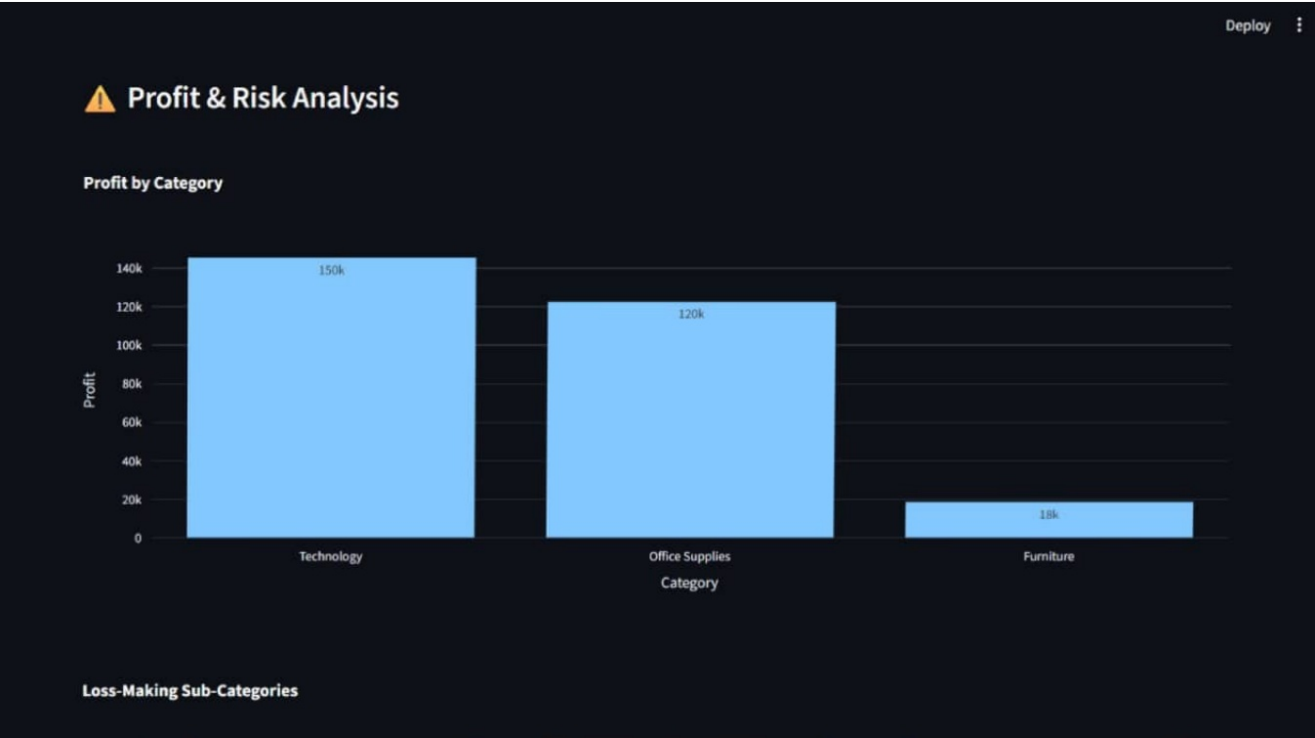


Figure A9: Profit and Risk Analysis – dashboard view 9.

Dashboard View 10: Profit and Risk Analysis

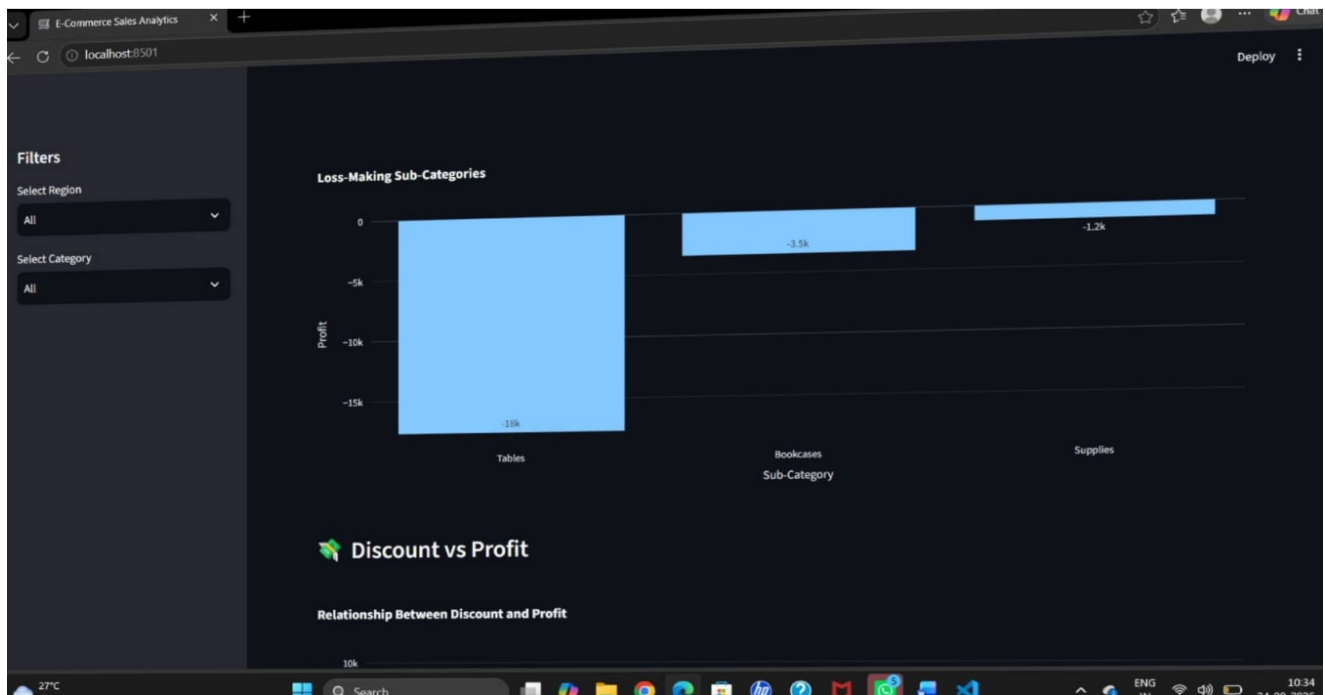


Figure A10: Profit and Risk Analysis – dashboard view 10.

Dashboard View 11: Profit and Risk Analysis

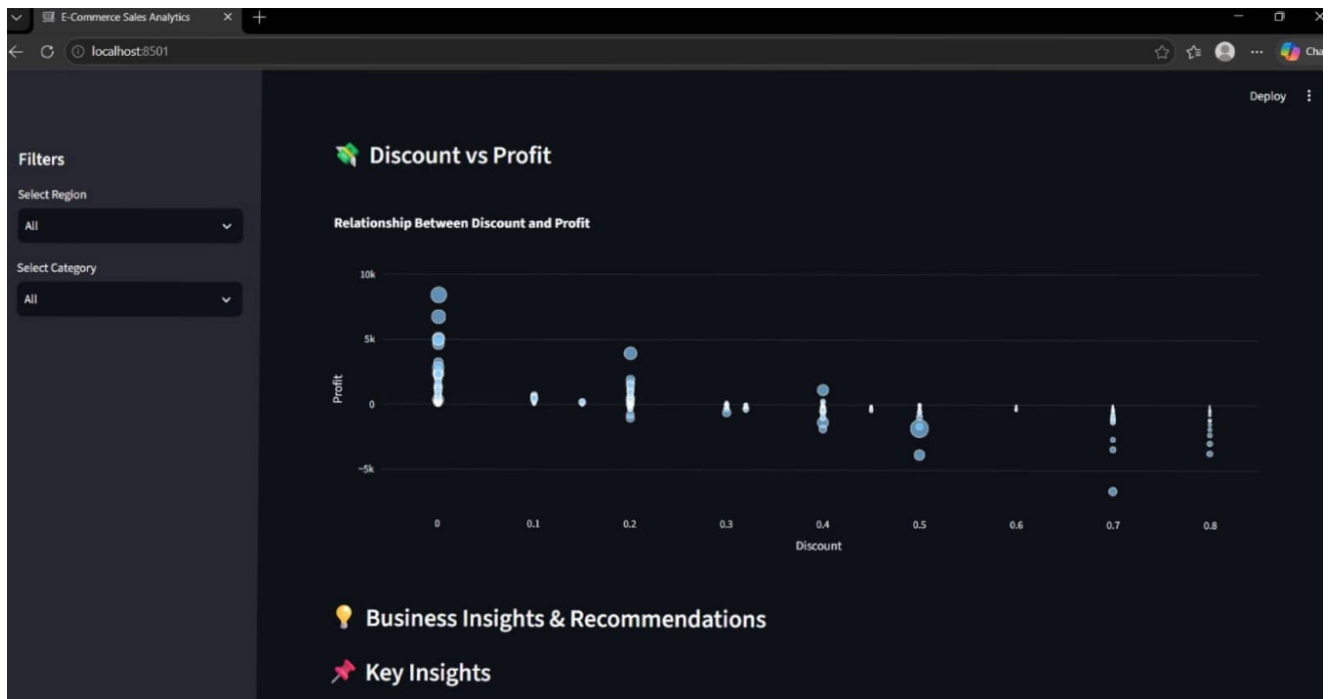


Figure A11: Profit and Risk Analysis – dashboard view 11.

Dashboard View 12: Business Insights and Recommendations

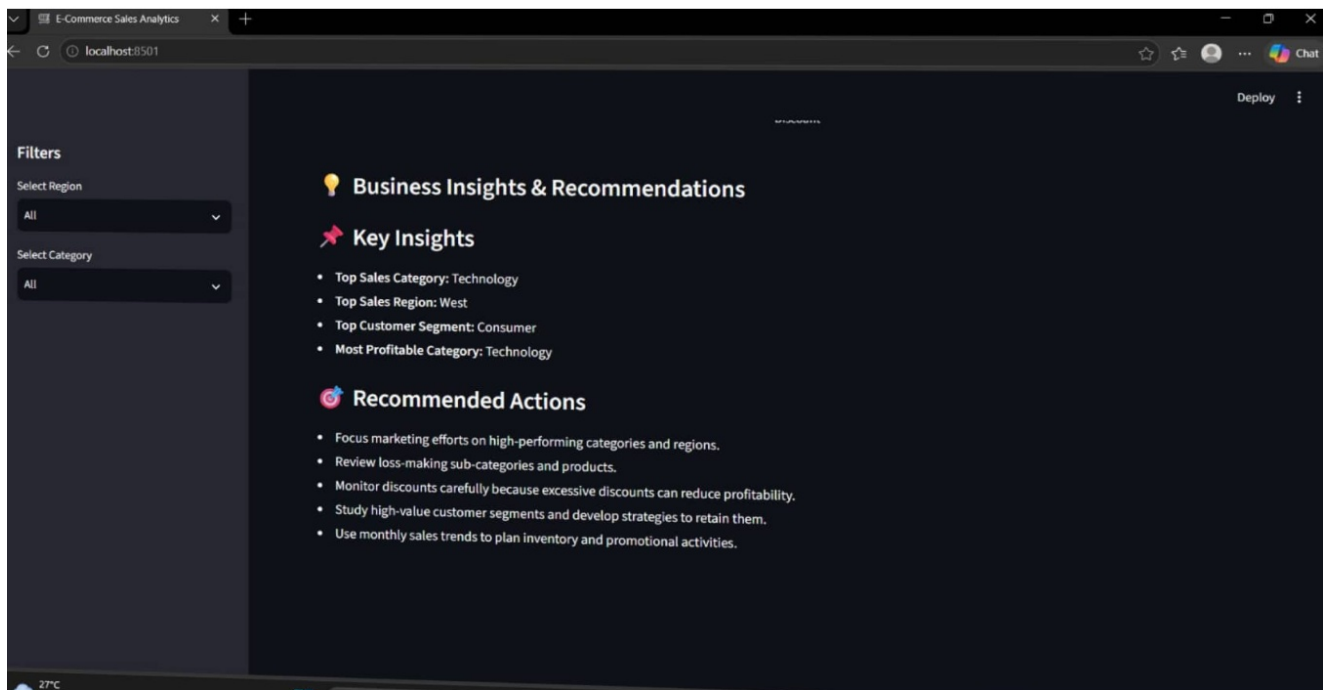


Figure A12: Business Insights and Recommendations – dashboard view 12.